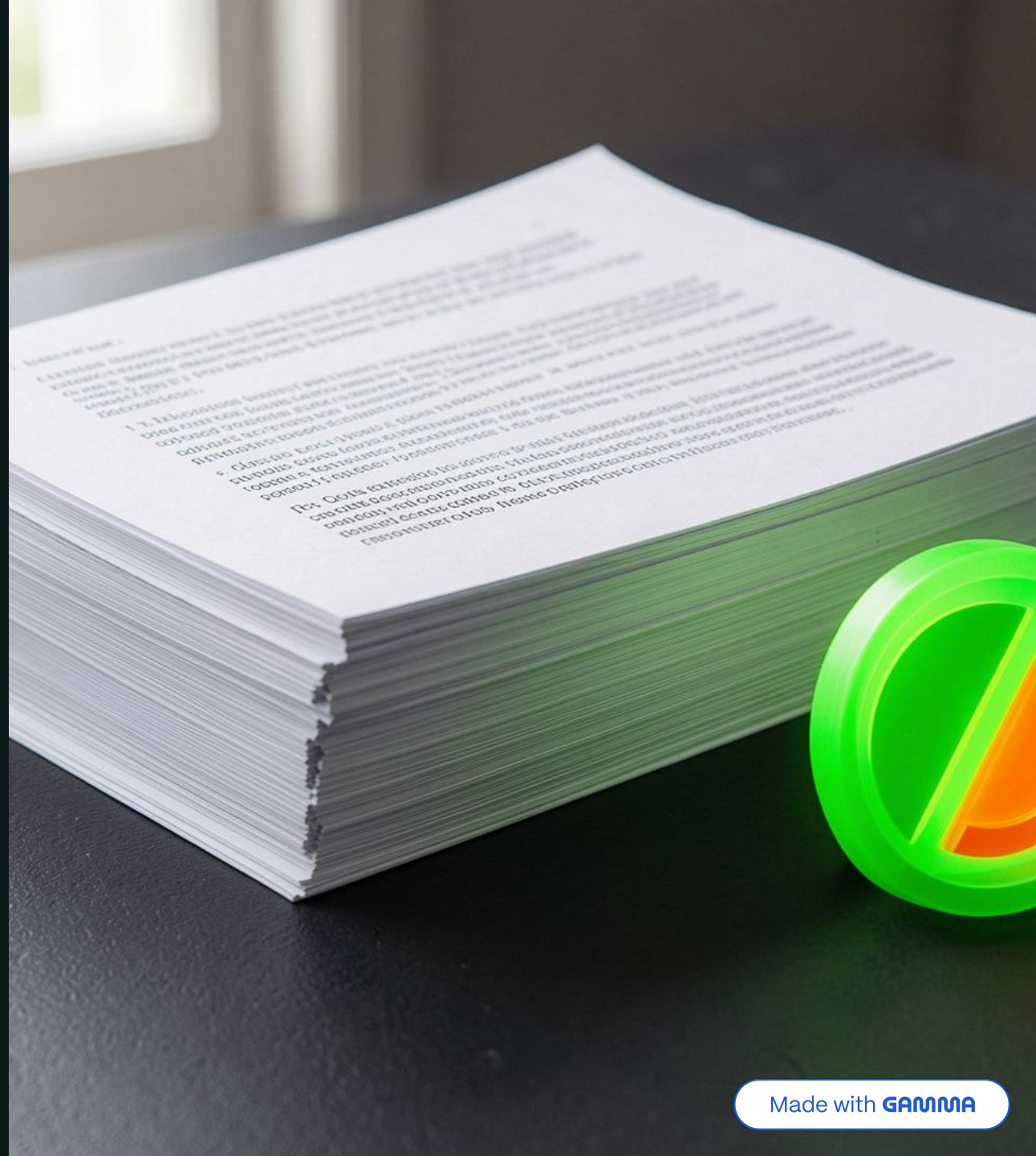
MACHINE-LEARNING PROJECT

LSTM-Based Sentiment Analysis

From review text to a reusable binary classifier.

IMDb reviews · TensorFlow / Keras · Google Colab



Made with GAMMA

From review text to an actionable sentiment label

A compact workflow with a clear output.

Build a binary classifier that predicts whether an IMDb movie review is positive or negative.

Run the complete workflow in **Google Colab** with **TensorFlow/Keras**.

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The problem is simple to state, hard to automate well

Movie reviews are unstructured text, making large-scale manual categorization slow and costly.

The model must learn meaningful language patterns, not just isolated keywords.

CORE CHALLENGE

Context matters.

Review language is a sequence, not a bag of disconnected words.

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The pipeline covers the full machine-learning lifecycle

1
Acquire and inspect
the dataset

2
Clean text
and encode labels

3
Tokenize and pad
reviews

4
Train and validate
the BiLSTM

5
Evaluate, predict,
and save reusable artifacts

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The dataset is IMDb reviews with binary labels

SOURCE

Zenodo record **7928582**, with a related Kaggle dataset page.

CORE FIELDS

Review text and sentiment label.

NOTE

Notebook filtering may reduce the final row count after cleaning and deduplication.

INPUT → LABEL

Review text

paired with

0 / 1

negative / positive

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Preprocessing turns messy language into model-ready sequences

The cleaning rules make the training input consistent before tokenization.

01

Clean the text

Remove HTML tags and normalize whitespace.

02

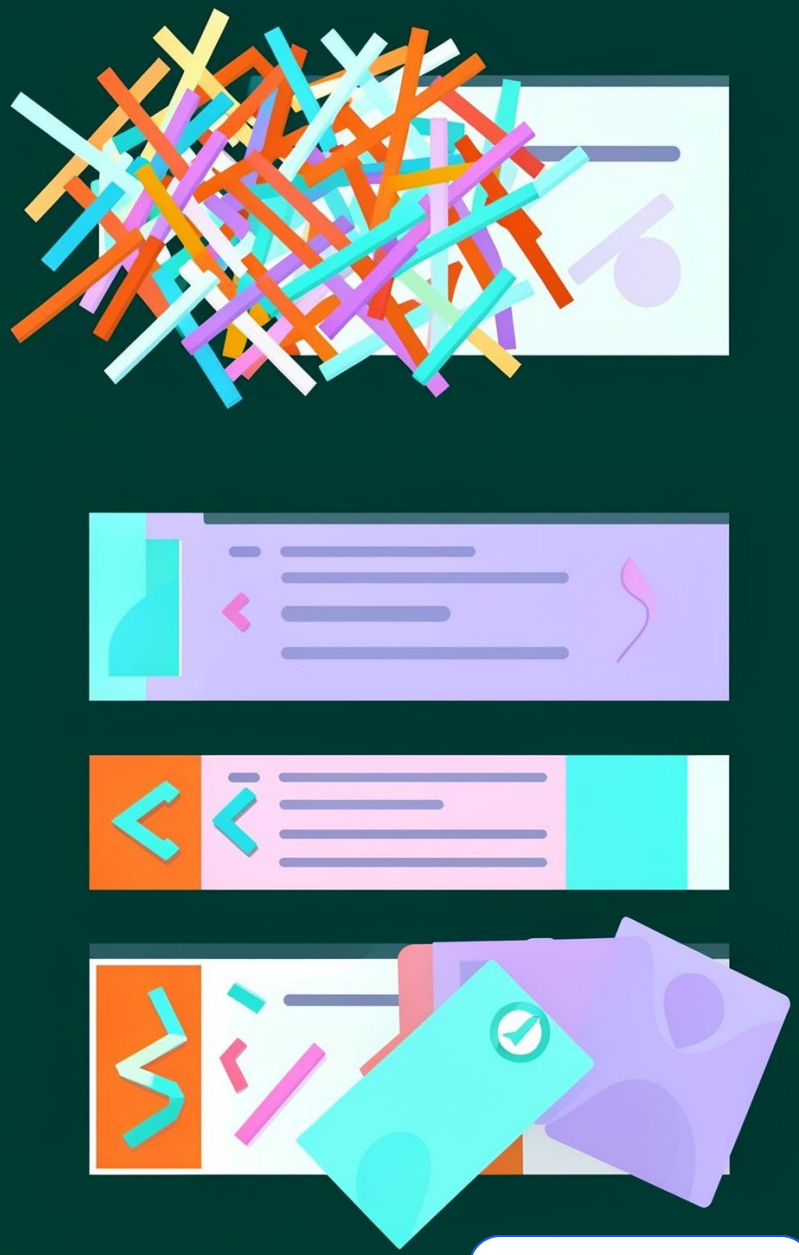
Filter the rows

Drop empty or unsupported rows and remove duplicate reviews.

03

Encode and split

Map negative to 0 and positive to 1; split 80% training and 20% test with stratification.



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Tokenization preserves a consistent input shape

Every review becomes a bounded sequence the network can read.

01

Vocabulary cap

30,000 words, with an out-of-vocabulary token included.

02

Fixed length

Each review is padded or truncated to **250 tokens**.

03

Leakage control

The tokenizer is fitted on training text only.

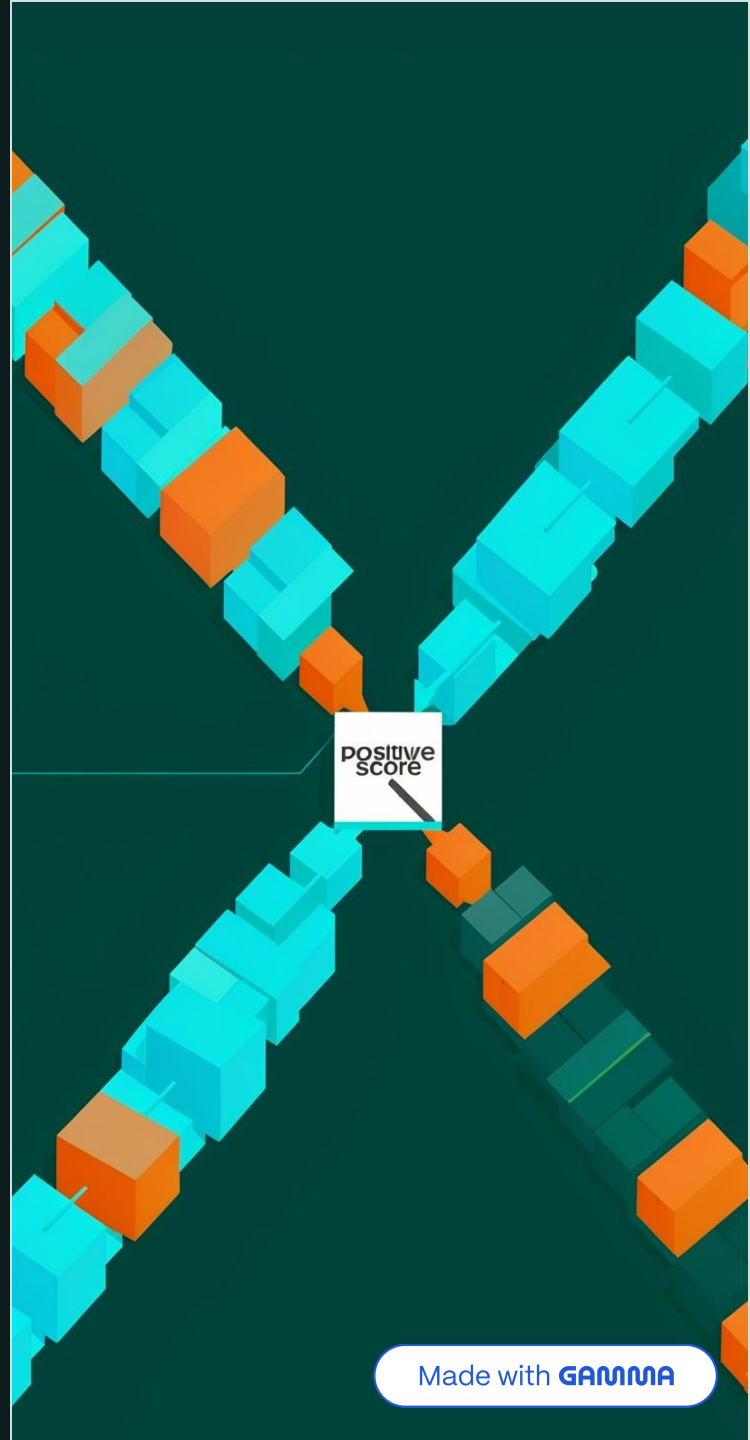
Text → token IDs → padded sequence of length 250

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Bidirectional context is the model's core advantage

The network combines token representations with context from both directions.

Embedding	Embedding layer learns dense token representations.
SpatialDropout1D	SpatialDropout1D regularizes across token features.
BiLSTM	Bidirectional LSTM reads sequence context in both directions.
Dense	Dense and dropout layers learn higher-level features while reducing overfitting.
Dropout	Sigmoid output produces a positive-class score from 0 to 1.
Sigmoid	



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Training is designed to stop when validation stops improving

Early stopping restores the best weights; learning rate is reduced when validation loss plateaus.

TRAINING SETUP

OPTIMIZER

Adam · learning rate **0.001**

OBJECTIVE

Binary cross-entropy loss

BUDGET

Up to **8 epochs** · batch size **128**

VALIDATION

15% validation split

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Evaluation measures more than accuracy

The notebook should report several views of model behavior, not one headline number.

Accuracy

Overall correctness.

Precision

Quality of positive predictions.

Recall

Positive cases captured.

F1-score

Balance of precision and recall.

Confusion matrix

False positives and false negatives.

Training curves

Convergence and possible overfitting.

Results caveat: measured results are not available until the notebook is run.

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